Local Economic Conditions and Airbnb Prices: Evidence from the New York City Housing Market

1. Project One

1.1 Introduction

The rapid expansion of the sharing economy has significantly transformed the tourism and accommodation industries. Among peer-to-peer accommodation platforms, Airbnb has become one of the most prominent examples, connecting homeowners who supply accommodation with travelers seeking short-term rentals. Since its founding in 2008, Airbnb has expanded rapidly across cities worldwide and has become an important component of urban tourism and local housing markets (Cai et al., 2019; Wang & Nicolau, 2017). As the platform continues to grow, understanding how prices are determined in short-term rental markets has become an increasingly important question for researchers and policymakers.

Previous studies have examined the factors that influence Airbnb listing prices. has examined the factors that influence Airbnb listing prices. Existing studies suggest that listing characteristics such as room type, property features, and amenities play an important role in shaping price variation because they reflect differences in accommodation quality (Wang & Nicolau, 2017; Cai et al., 2019). In addition, host characteristics and online reputation signals, including ratings and the number of reviews, can influence consumer perceptions and willingness to pay by reducing information asymmetry in online markets (Lin & Yang, 2024).

Beyond listing characteristics, location and neighborhood conditions are also important determinants of Airbnb prices. Similar to traditional housing and hotel markets, prices in the short-term rental market often reflect spatial differences within cities. Listings closer to tourist attractions, transportation hubs, or central areas typically command higher prices because of stronger tourism demand and better accessibility (Chica-Olmo et al., 2020). However, pricing patterns may also vary across neighborhoods depending on local amenities, accessibility, and surrounding urban environments (Perez-Sanchez et al., 2018; Jiang et al., 2023). Spatial interactions between nearby listings may further influence price patterns across urban areas (Gyodi & Nawaro, 2021).

At the same time, Airbnb pricing is closely connected to broader urban housing market conditions. Short-term rental platforms interact with local housing markets and may influence housing supply, tourism demand, and neighborhood development patterns (Zhu & Liu, 2024). Differences in neighborhood economic conditions and urban environments can therefore affect the attractiveness of different areas for visitors and shape short-term rental prices (Wind & Ibrahim, 2020).

Motivated by these considerations, this project investigates the following research question: ***how neighborhood-level economic conditions are associated with Airbnb price levels through housing market conditions***. To explore this relationship, the analysis uses the *NYC Airbnb 2019* dataset, which provides detailed information on listing prices, housing characteristics, and spatial locations across New York City. These data are combined with borough-level economic indicators and neighborhood crime data to examine how local economic environments and housing market conditions relate to short-term rental prices. The study focuses on three key dimensions: spatial differences across boroughs, housing characteristics of listings, and neighborhood safety conditions.

By combining descriptive spatial analysis with regression analysis, this project highlights how neighborhood economic environments and housing market conditions are reflected in Airbnb prices and provides insights into the broader economic implications of short-term rental platforms for urban housing markets.

1.2 Data Description & Cleaning

1.2.1 Original (Main) Dataset

The primary dataset is the AB_NYC_2019 Airbnb dataset, which contains listing-level information for New York City in 2019. Each observation represents an individual listing and includes variables related to price, location,

room type, and availability. The listing price is used as the main outcome variable. To reduce noise, variables with substantial missing values and limited relevance to neighborhood-level economic mechanisms, such as **last_review** and **reviews_per_month**, are dropped. The dataset is later merged with borough-level economic data using the neighbourhood_group identifier.

1.2.2 Other Dataset

To capture neighborhood-level economic and social conditions, several external datasets are incorporated and merged with the Airbnb data.The first dataset is sourced from the 2020 U.S. Census Data Report and provides borough-level information for New York City, including population and land area.These variables are used to construct population density and GDP per capita, which proxy local economic activity. The second dataset, obtained from Tableau Public, provides borough-level average household income in 2019.In addition, crime data from the New York City Police Department are incorporated to capture neighborhood safety conditions. Therefore airbnb listings are matched to police precincts using geographic coordinates.

1.2.3 Variable Choice

The outcome variable is the Airbnb listing price, and the key explanatory variables include borough-level economic indicators such as average household income, GDP per capita, population density and crime rate. At the same time, listing-level market characteristics,such as the number of total reviews, minimum nights and availability (measured by the number of available days) is used to proxy short-term rental supply tightness and listing activity, while room type is included as a categorical variable to capture differences in housing product composition across neighborhoods. The underlying mechanism is that neighborhood economic conditions shape Airbnb prices through both demand-side willingness to pay and supply side housing market constraints. Higher-income and more economically productive neighborhoods are expected to face stronger demand, while differences in room type composition and availability reflect variation in effective housing supply for short-term rentals. Together, these factors help explain systematic differences in Airbnb prices across New York City.

1.3 Summary Statistics

1.3.1 Main Variables

In [2]:
```
# Import Datasets and Cleaning
import pandas as pd
import numpy as np

df = pd.read_csv("AB_NYC_2019.csv")

# Import Basic NYC Neighbourhood-level Economic Indicators
nbh_data = pd.DataFrame({
    'neighbourhood_group':['Bronx', 'Brooklyn','Manhattan', 'Queens', 'Staten Island'],
    'Population':[1472654, 2736074, 1694251, 2405464, 495747],
    'Area':[109.2, 179.7, 58.7, 281.6, 149.0],
    'GDP_in_billion':[51.574, 125.867, 885.652, 122.288, 21.103]
    })

nbh_data['Pop_Density'] = nbh_data['Population'] / nbh_data['Area']
nbh_data['GDP_per_capita'] = nbh_data['GDP_in_billion'] / nbh_data["Population"] * 1e9
```

In [3]:
```
# Import Neighbourhood Avg Income Data
Avg_Income = pd.read_csv("Average Households Income Across the Regions.csv")

Avg_Income = Avg_Income.rename(columns={
    "Householdsmean Income Dollars": "Avg Income",
    "Geographic Area Name": "neighbourhood_group"})

Avg_Income_2019 = Avg_Income.query('Year == 2019').copy()

Avg_Income_2019["neighbourhood_group"] = Avg_Income_2019["neighbourhood_group"].replace({
    "Kings County, New York": "Brooklyn",
    "Queens County, New York": "Queens",
    "Richmond County, New York": "Staten Island",
    "New York County, New York": "Manhattan",
    "Bronx County, New York": "Bronx"})
```

```
Avg_Income_2019 = Avg_Income_2019[["neighbourhood_group", "Avg Income"]]
nbh_data_m = pd.merge(nbh_data, Avg_Income_2019, on='neighbourhood_group')
```

In [4]:
```
# Import Crime Dataset
import geopandas as gpd
cd = pd.read_csv("seven-major-felony-offenses-by-precinct-2000-2025.csv")

cd["PCT"] = cd["PCT"].ffill()
total_cd = cd[cd["CRIME"].str.contains("TOTAL", case=False, na=False)]
crime_2019 = total_cd[["PCT", "Count_Offences"]]

airbnb_geo = gpd.GeoDataFrame(
    df,
    geometry=gpd.points_from_xy(df.longitude, df.latitude),
    crs="EPSG:4326")

precincts = gpd.read_file("nyc_precincts.gpkg").to_crs("EPSG:4326")
```

In [5]:
```
# Data Cleaning

airbnb_pct = gpd.sjoin(airbnb_geo, precincts[["Precinct", "geometry"]], how="left", predicate="within")

# Extract only listing id + precinct
pct_match = airbnb_pct[["id", "Precinct"]].copy()
pct_match["Precinct"] = pd.to_numeric(pct_match["Precinct"], errors="coerce")

crime_2019 = crime_2019.copy()
crime_2019["PCT"] = pd.to_numeric(crime_2019["PCT"], errors="coerce")

df = df.merge(pct_match, on="id", how="left")
df = df.merge(crime_2019, left_on="Precinct", right_on="PCT", how="left")

df["Count_Offences"] = (
    df["Count_Offences"]
    .astype(str)
    .str.replace(",", "")
    .astype(float)
)

# Calculate crime rate (>1 means higher than avg, <1 means lower than avg)
df["crime_rate"] = df["Count_Offences"] / df["Count_Offences"].mean()

df = df.drop(columns=["last_review","reviews_per_month","Precinct"])

# Merge original dataframe & economic indicator data
df1 = df.merge(nbh_data_m, on="neighbourhood_group", how="left")

df1['Price_to_Income Ratio'] = df1['price']*365 / (df1['Avg Income'])

df1["log_price"] = np.log(df1["price"] + 1)

df1["avg_income_10k"] = df1["Avg Income"] / 10000
df1["gdp_10k"] = df1["GDP_per_capita"] / 10000
df1["density_1k"] = df1["Pop_Density"] / 1000
```

In [6]:
```
from IPython.display import display, Latex
from stargazer.stargazer import Stargazer
from IPython.core.display import HTML
```

In [24]:
```
# ==============================
# Table 1: Main Quantity Variables Summary
# ==============================

vars_table1 = [
    "price",
    "Avg Income",
    "GDP_per_capita",
    "Pop_Density",
    "Count_Offences",
    "availability_365",
    "minimum_nights",
    "number_of_reviews"]

tbl1 = df1[vars_table1].replace([np.inf, -np.inf], np.nan)
tbl1["price_income_ratio"] = tbl1["price"] * 365 / tbl1["Avg Income"]
```

```
summary = tbl1.agg(["count","mean","median","std","min","max"]).T
summary.columns = ["N","Mean","Median","SD","Min","Max"]

summary["N"] = summary["N"].astype(int)
summary[["Mean","Median","SD","Min","Max"]] = summary[["Mean","Median","SD","Min","Max"]].round(2)

summary = summary.reset_index().rename(columns={"index":"Statistic"})

name_map = {
    "price": "Price",
    "Avg Income": "AverageIncome",
    "GDP_per_capita": "GDPperCapita",
    "Pop_Density": "PopulationDensity",
    "Count_Offences": "Crime Counts",
    "availability_365": "Availability (365 days)",
    "minimum_nights": "MinimumNights",
    "number_of_reviews": "Reviews",
    "price_income_ratio": "Price To Income Ratio"}

summary["Statistic"] = summary["Statistic"].replace(name_map)

rows = []
for _, r in summary.iterrows():
    rows.append(
        f"{r['Statistic']} & "
        f"{int(r['N'])} & "
        f"{r['Mean']:.2f} & "
        f"{r['Median']:.2f} & "
        f"{r['SD']:.2f} & "
        f"{r['Min']:.2f} & "
        f"{r['Max']:.2f} \\\\"
    )

latex = r"""
\[
\begin{array}{lrrrrrr}
\textbf{Statistic} & \textbf{N} & \textbf{Mean} & \textbf{Median} & \textbf{SD} & \textbf{Min} & \textbf{Ma
\hline
""" + "\n".join(rows) + r"""
\end{array}
\]
"""

display(Latex(latex))
```

Statistic	N	Mean	Median	SD	Min	Max
Price	48895	152.72	106.00	240.15	0.00	10000.00
AverageIncome	48895	129804.94	99472.00	36738.30	59664.00	170453.00
GDPperCapita	48895	257491.21	50837.59	236570.92	35021.13	522739.55
PopulationDensity	48895	20363.06	15225.79	7894.34	3327.16	28862.88
CrimeCounts	48895	1329.53	1234.00	510.06	205.00	3525.00
Availability(365*days*)	48895	112.78	45.00	131.62	0.00	365.00
MinimumNights	48895	7.03	3.00	20.51	1.00	1250.00
Reviews	48895	23.27	5.00	44.55	0.00	629.00
PriceToIncomeRatio	48895	0.43	0.32	0.65	0.00	37.77

Table 1 shows that Airbnb prices are highly dispersed and right-skewed, with the mean substantially exceeding the median. This indicates a distribution driven by a small number of high-priced outliers. Furthermore, neighborhood-level variables—such as income, population density, and crime rates—vary considerably across boroughs, reflecting substantial spatial heterogeneity in local economic conditions. These patterns collectively motivate the subgroup analysis that follows.

1.3.2 Categorical Variables

In [26]:
```
# ==============================
# Table 2.1: Counts by Borough and Room Type
# ==============================
```

```
count_table = (
    df1.groupby(["neighbourhood_group", "room_type"])
    .size()
    .unstack(fill_value=0)
)

desired_cols = ["Entire home/apt", "Private room", "Shared room"]
count_table = count_table[[c for c in desired_cols if c in count_table.columns]]
count_table["Total"] = count_table.sum(axis=1)

count_table.index = count_table.index.str.replace("_", "")

rows = []
for borough, r in count_table.iterrows():
    rows.append(
        f"{borough} & "
        f"{int(r['Entire home/apt'])} & "
        f"{int(r['Private room'])} & "
        f"{int(r['Shared room'])} & "
        f"{int(r['Total'])} \\\\"
    )

latex = r"""

\[
\begin{array}{lrrrr}
\textbf{Borough} & \textbf{Entire home/apt} & \textbf{Private room} & \textbf{Shared room} & \textbf{Total}
\hline
""" + "\n".join(rows) + r"""
\end{array}
\]
"""

display(Latex(latex))
```

Borough	Entire home/apt	Private room	Shared room	Total
Bronx	379	652	60	1091
Brooklyn	9559	10132	413	20104
Manhattan	13199	7982	480	21661
Queens	2096	3372	198	5666
StatenIsland	176	188	9	373

Table 2 shows clear differences in room type composition across boroughs. Manhattan has more entire homes and apartments, while the Bronx and Queens are dominated by private rooms. Shared rooms remain rare across all boroughs. This heterogeneity reflects systematic differences in short-term rental supply across neighborhoods and motivates the subgroup analysis that follows.

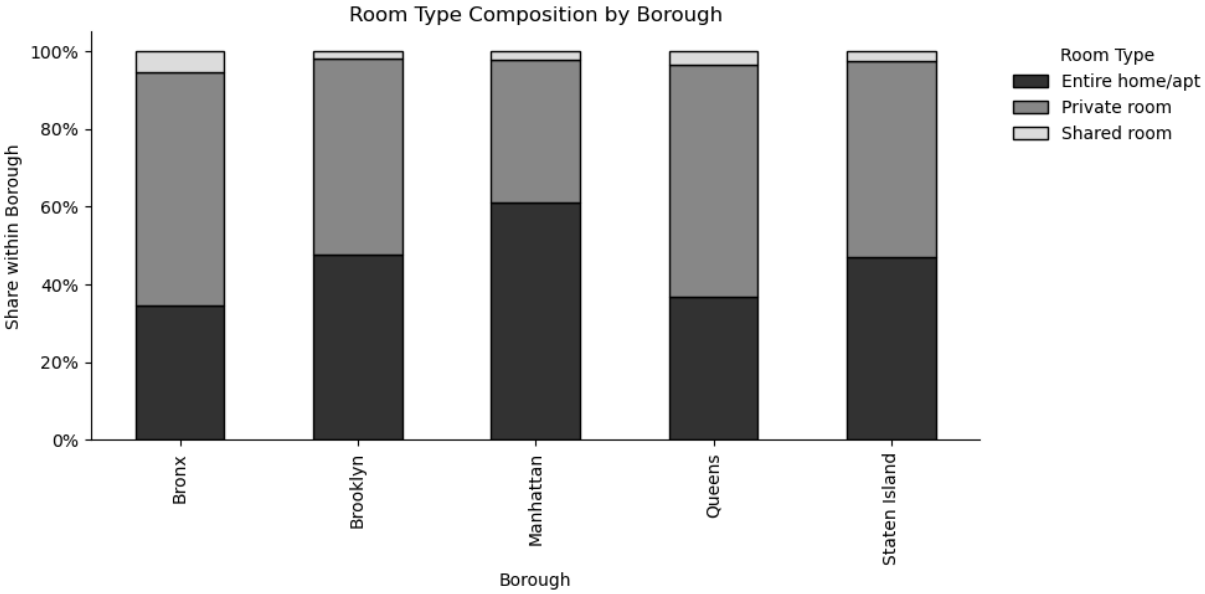
1.4 Plots

1.4.1 Figure 1: Room Type Distribution Among Different Neighbourhood

In [28]:
```
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
```

In [29]:
```
#==============================================
# Figure 1: Room Type Distribution Comparison
#==============================================

counts = pd.crosstab(df["neighbourhood_group"], df["room_type"])
shares = counts.div(counts.sum(axis=1), axis=0)

colors = ["#333333","#888888","#DDDDDD"]

ax = shares.plot(
    kind="bar",
    stacked=True,
    figsize=(10, 5),
    color=colors,
    edgecolor="black")
```

```
ax.set_ylabel("Share within Borough")
ax.set_xlabel("Borough")
ax.set_title("Room Type Composition by Borough")
ax.yaxis.set_major_formatter(mtick.PercentFormatter(1.0))

ax.legend(
    title="Room Type",
    bbox_to_anchor=(1.02, 1),
    loc="upper left",
    frameon=False)

ax.spines['right'].set_visible(False)
ax.spines['top'].set_visible(False)

plt.tight_layout()
plt.show()
```

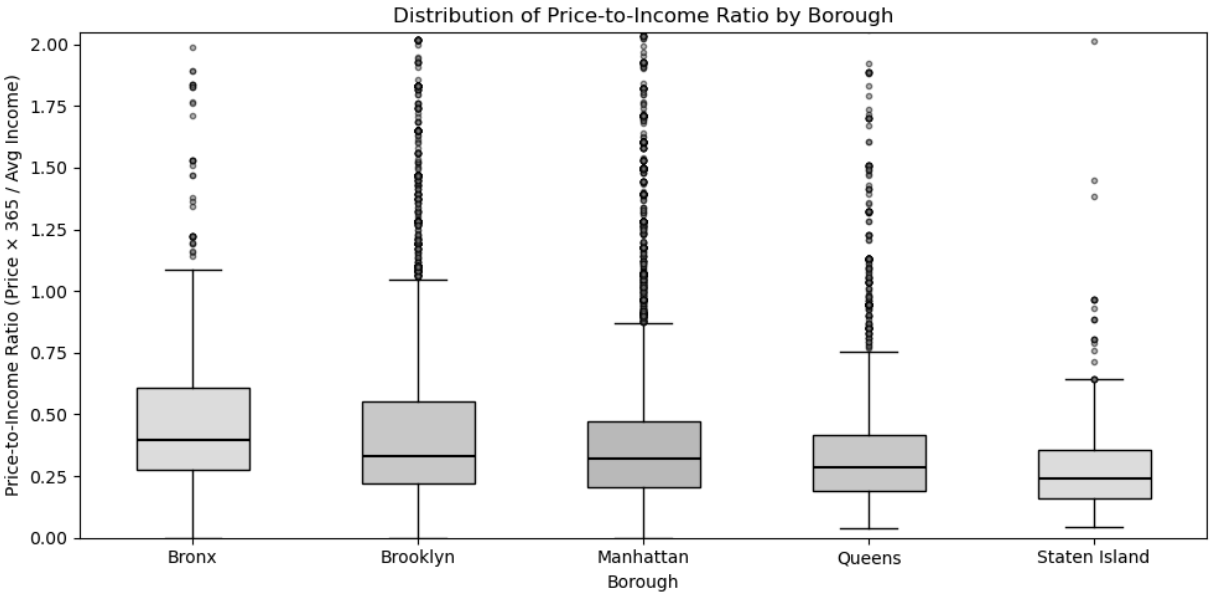


This figure uses room type to describe the composition of Airbnb supply across boroughs. Room type matters because entire homes and private rooms represent different housing products and are usually priced differently. The figure asks whether the structure of Airbnb supply varies across boroughs. The plot shows that Manhattan has a much higher share of entire-home listings, while the Bronx and Queens are dominated by private rooms. This suggests that Airbnb market structure differs systematically across boroughs and may contribute to observed price differences.

1.4.2 Figure 2: Affordability Comparison through Price-to-income Ratio

In [5]:
```
# ==========================================
# Figure 2: Price-to-income Ratio Comparison
# ==========================================

borough_order = ["Bronx", "Brooklyn", "Manhattan", "Queens", "Staten Island"]

data = []
labels = []

for b in borough_order:
    sub = (
        df1.loc[df1["neighbourhood_group"] == b, "Price_to_Income Ratio"]
           .replace([np.inf, -np.inf], np.nan)
           .dropna())
    if len(sub) > 0:
        data.append(sub.to_numpy())
        labels.append(b)

fig, ax = plt.subplots(figsize=(10, 5))

bp = ax.boxplot(
    data,
    showfliers=True,
```

```
    patch_artist=True,
    medianprops=dict(color="black", linewidth=1.5),
    boxprops=dict(edgecolor="black"),
    whiskerprops=dict(color="black"),
    capprops=dict(color="black"),
    flierprops=dict(marker="o", markersize=3, markerfacecolor="gray", alpha=0.5)
)

fills = ["#DDDDDD", "#CCCCCC", "#BBBBBB", "#CCCCCC", "#DDDDDD"]
for patch, c in zip(bp["boxes"], fills):
    patch.set_facecolor(c)

ax.set_xticks(range(1, len(labels) + 1))
ax.set_xticklabels(labels)

ax.set_title("Distribution of Price-to-Income Ratio by Borough")
ax.set_xlabel("Borough")
ax.set_ylabel("Price-to-Income Ratio (Price × 365 / Avg Income)")

upper = np.nanpercentile(df1["Price_to_Income Ratio"], 99)
ax.set_ylim(0, upper)

plt.tight_layout()
plt.show()
```

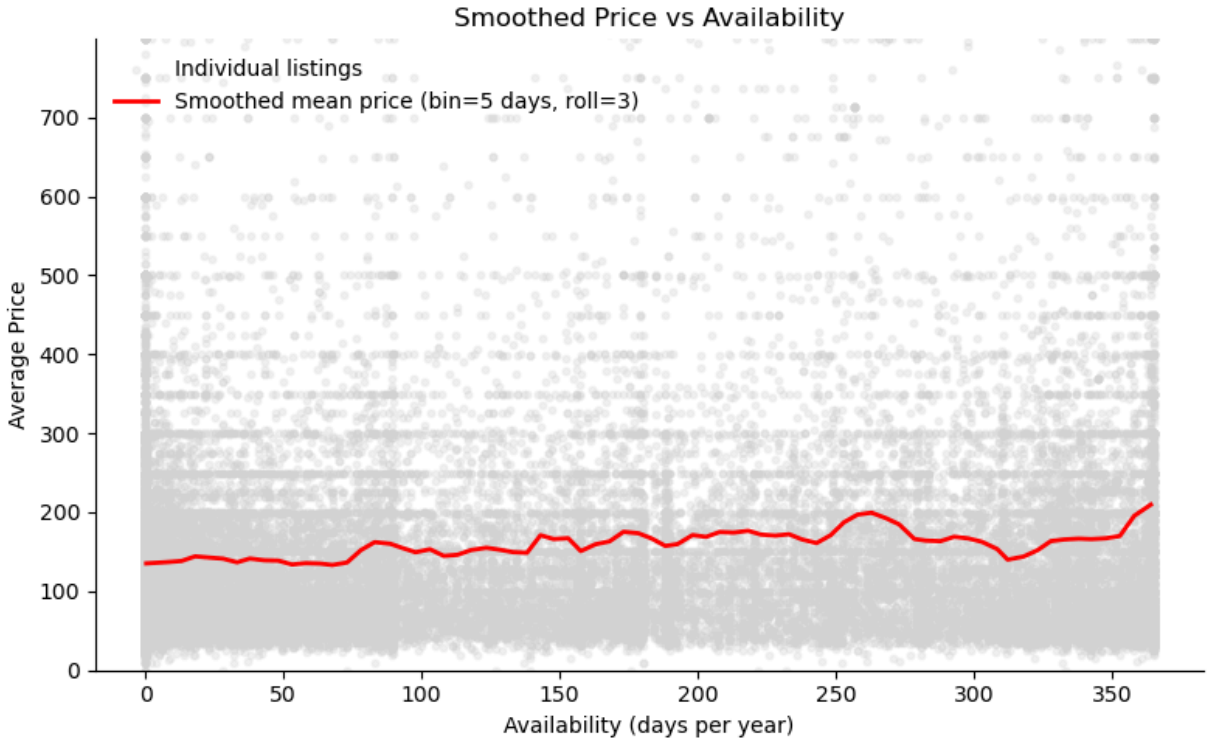


This figure uses the price-to-income ratio to measure Airbnb affordability across boroughs. This variable is chosen because it scales prices by local income, making comparisons across locations more meaningful. The figure asks whether Airbnb listings are similarly affordable in different boroughs. The boxplots show that the median price-to-income ratio is highest in the Bronx and lowest in Staten Island, with substantial variation across boroughs. This indicates that Airbnb affordability differs significantly by location.

1.4.3 Figure 3: Relationship between Price & Availability

In [6]:
```
# ==================================
# Figure 3: Price VS. Availability
# ==================================

dfp = df1[["availability_365", "price"]].copy()
dfp = dfp.replace([np.inf, -np.inf], np.nan).dropna()

fig, ax = plt.subplots(figsize=(8, 5))
ax.scatter(
    dfp["availability_365"],
    dfp["price"],
    color="lightgray",
    alpha=0.25,
    s=10,
    label="Individual listings")

bin_width = 5
bins = np.arange(0, 365 + bin_width, bin_width)
```

```
dfp["avail_bin"] = pd.cut(dfp["availability_365"], bins=bins, include_lowest=True)

binned = (
    dfp.groupby("avail_bin", observed=True)
       .agg(mean_avail=("availability_365", "mean"),
            mean_price=("price", "mean"),
            n=("price", "size"))
       .dropna()
       .sort_values("mean_avail")
)

window = 3
binned["smooth_price"] = binned["mean_price"].rolling(window=window, center=True, min_periods=1).mean()

ax.plot(
    binned["mean_avail"],
    binned["smooth_price"],
    color="red",
    linewidth=2,
    label=f"Smoothed mean price (bin={bin_width} days, roll={window})"
)

ax.set_ylim(0, np.percentile(dfp["price"], 99))
ax.set_title("Smoothed Price vs Availability")
ax.set_xlabel("Availability (days per year)")
ax.set_ylabel("Average Price")
ax.legend(frameon=False)
ax.spines['right'].set_visible(False)
ax.spines['top'].set_visible(False)

plt.tight_layout()
plt.show()
```

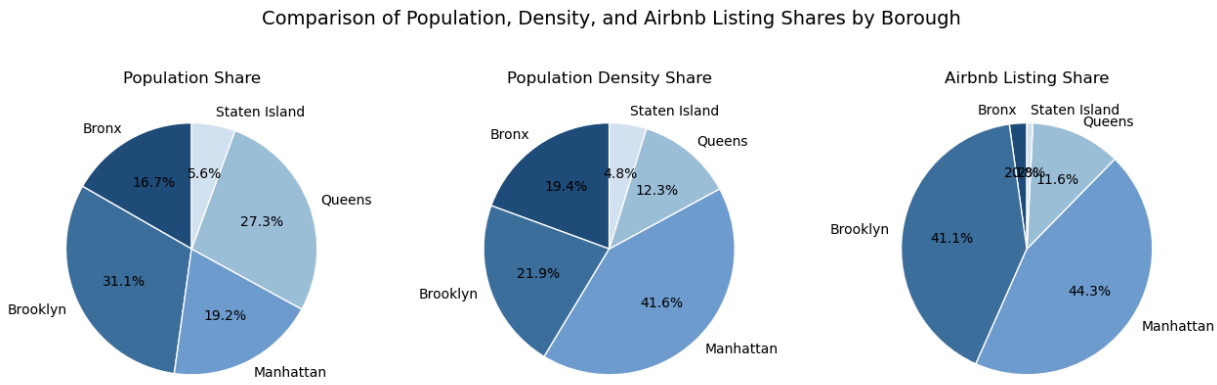


This figure examines the relationship between Airbnb prices and availability. Availability is chosen because it reflects how often a listing is available for booking and serves as a proxy for supply conditions. The economic question is whether prices seen to vary systematically with availability. While individual prices are highly dispersed, the smoothed average line is relatively flat. This suggests that availability alone does not strongly explain Airbnb price differences at the aggregate level.

1.4.4 Figure 4: Population VS Listing Share

In [7]:
```
# ========================================
# Figure 4: Population VS Listing Share
# ========================================
```

```python
listing_count = (df1.groupby("neighbourhood_group")
                  .size()
                  .reset_index(name="listings"))

pop_density = (
    nbh_data_m[
        ["neighbourhood_group", "Population", "Pop_Density"]
    ]
    .drop_duplicates())

dfp2 = (listing_count.merge(pop_density, on="neighbourhood_group", how="left"))

# Calculate Shares
dfp2["listing_share"] = dfp2["listings"] / dfp2["listings"].sum()
dfp2["population_share"] = dfp2["Population"] / dfp2["Population"].sum()
dfp2["density_share"] = dfp2["Pop_Density"] / dfp2["Pop_Density"].sum()

dfp2["neighbourhood_group"] = pd.Categorical(
    dfp2["neighbourhood_group"],
    categories=borough_order,
    ordered=True)
dfp2 = dfp2.sort_values("neighbourhood_group")

colors = ["#1F4E79","#3C6E9C","#6F9FCF","#9FBFDB","#D6E4F0"]
labels = dfp2["neighbourhood_group"]

fig, axes = plt.subplots(1, 3, figsize=(13, 4))

# Pie1: Population Share
axes[0].pie(
    dfp2["population_share"],
    labels=labels,
    autopct="%1.1f%%",
    startangle=90,
    colors=colors,
    wedgeprops=dict(edgecolor="white")
)
axes[0].set_title("Population Share")

# Pie2: Population Density Share
axes[1].pie(
    dfp2["density_share"],
    labels=labels,
    autopct="%1.1f%%",
    startangle=90,
    colors=colors,
    wedgeprops=dict(edgecolor="white")
)
axes[1].set_title("Population Density Share")

# Pie3: Airbnb Listing Share
axes[2].pie(
    dfp2["listing_share"],
    labels=labels,
    autopct="%1.1f%%",
    startangle=90,
    colors=colors,
    wedgeprops=dict(edgecolor="white")
)
axes[2].set_title("Airbnb Listing Share")

fig.suptitle(
    "Comparison of Population, Density, and Airbnb Listing Shares by Borough",
    fontsize=14,
    y=1.05
)

plt.tight_layout()
plt.show()
```



This figure compares borough-level shares of population, density, and Airbnb listings to examine whether supply aligns with local demographic patterns. The data reveal a disproportionate concentration of short-term rentals in Manhattan and Brooklyn, where listing shares far exceed population shares.

Integrating Figures 2 and 4 highlights a decoupling between listing density and affordability: this spatial concentration does not necessarily translate to lower prices relative to local income. In fact, some boroughs with fewer listings maintain higher price-to-income ratios. A potential explanation lies in neighborhood safety; higher crime rates may suppress visitor willingness to pay, thereby shaping pricing outcomes. We explore this spatial relationship between crime and pricing in the subsequent mapping section.

2. Project Two

2.1 The Message

Message

Airbnb listings located in neighborhoods with stronger economic environments, reflected in higher income levels, tend to exhibit higher price levels.

Note: In this study, the neighborhood economic environment refers broadly to local economic and social conditions, which influence local housing demand and supply. These conditions include income levels, economic productivity(proxied by GDP per capita), population density, and neighborhood safety (proxied by crime rates). Average income level serves as the primary observable indicator of local economic strength in the boxplot visualization presented below.

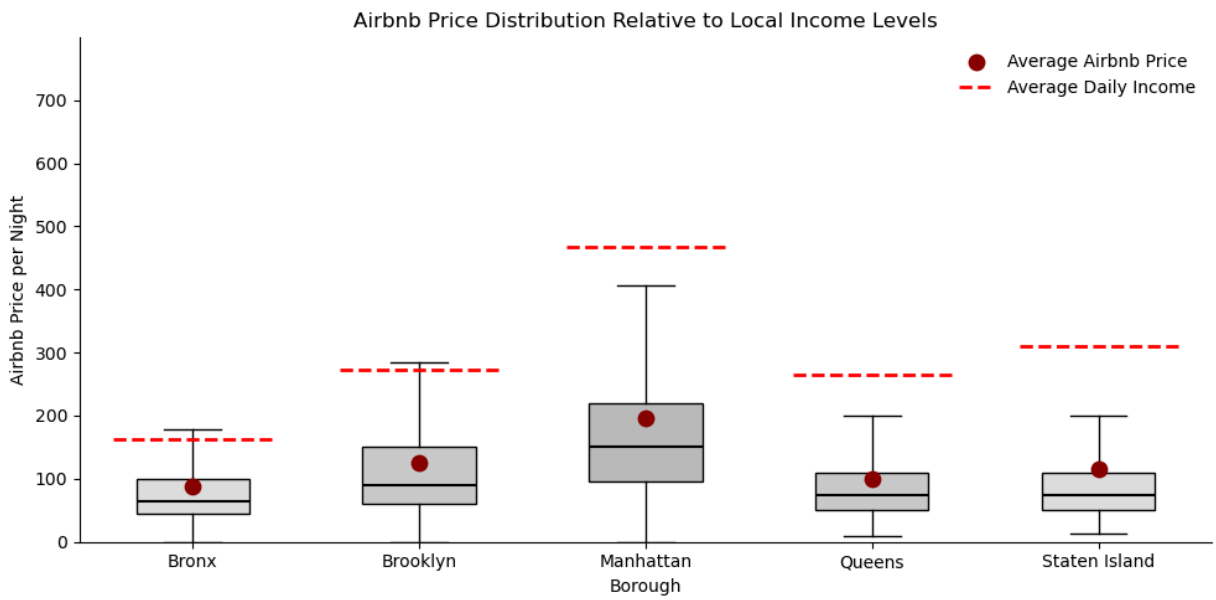
Description

The figure illustrates the distribution of Airbnb listing prices across the boroughs of New York City, and compares them with the average daily household income levels in each borough. The boxplots display the distribution of Airbnb listing prices, while the dashed red lines represent the average daily income level, calculated as annual household income divided by 365.

A positive association between local income levels and Airbnb prices can be observed. Boroughs with stronger economic conditions, such as Manhattan, tend to exhibit substantially higher Airbnb prices, whereas boroughs with lower income levels display lower price distributions. This pattern provides intuitive evidence that differences in neighborhood economic environments are associated with systematic variation in Airbnb price levels.

In [10]:
```
# Data Measurement
borough_order = ["Bronx", "Brooklyn", "Manhattan", "Queens", "Staten Island"]
price_clean = df1["price"].replace([np.inf, -np.inf], np.nan).dropna()

upper = np.percentile(price_clean, 99)

data = []
mean_prices = []
daily_income = []

for b in borough_order:
    sub = df1.loc[df1["neighbourhood_group"] == b, "price"]
```

```python
    sub = sub.replace([np.inf, -np.inf], np.nan).dropna()
    data.append(sub)
    mean_prices.append(sub.mean())
    income = (
        df1.loc[df1["neighbourhood_group"] == b, "Avg Income"]
        .replace([np.inf, -np.inf], np.nan)
        .dropna()
        .mean() / 365)
    daily_income.append(income)

# Create figure
fig, ax = plt.subplots(figsize=(10,5))

bp = ax.boxplot(
    data,
    showfliers=False,
    patch_artist=True,
    medianprops=dict(color="black", linewidth=1.5),
    boxprops=dict(edgecolor="black"),
    whiskerprops=dict(color="black"),
    capprops=dict(color="black")
)

colors = ["#DDDDDD","#CCCCCC","#BBBBBB","#CCCCCC","#DDDDDD"]

for patch, c in zip(bp["boxes"], colors):
    patch.set_facecolor(c)

x = np.arange(1, len(borough_order)+1)

ax.scatter(x, mean_prices, color="darkred", s=80, zorder=3, label="Average Airbnb Price")

# Daily income dashed lines
for i, inc in enumerate(daily_income):
    ax.hlines(
        inc,
        i+1-0.35,
        i+1+0.35,
        colors="red",
        linestyles="dashed",
        linewidth=2,
        label="Average Daily Income" if i == 0 else ""
    )

# Plot Labels
ax.set_xticks(x)
ax.set_xticklabels(borough_order)

ax.set_ylabel("Airbnb Price per Night")
ax.set_xlabel("Borough")
ax.set_title("Airbnb Price Distribution Relative to Local Income Levels")

# Limit extreme outliers
ax.set_ylim(0, upper)
ax.legend(frameon=False)

ax.spines['right'].set_visible(False)
ax.spines['top'].set_visible(False)

plt.tight_layout()
plt.show()
```



2.2 Maps and Interpretations

To better understand the spatial distribution of Airbnb prices and neighborhood conditions across New York City, this section presents a series of maps at the ZIP code level. These maps visualize key variables related to the neighborhood economic environment and the Airbnb market, providing intuitive insights into how neighborhood characteristics vary across the city and how they may be associated with differences in Airbnb price levels.

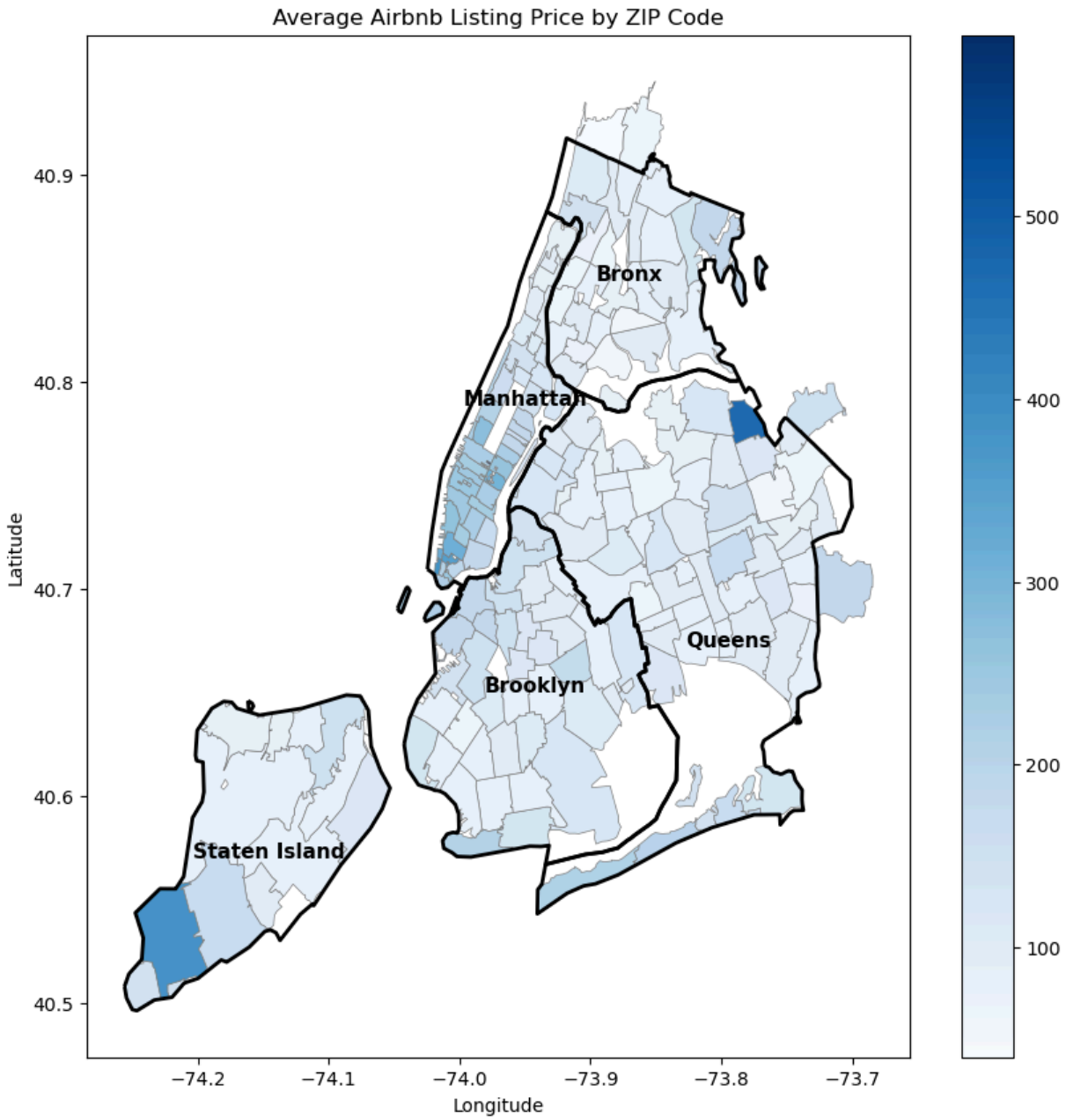
In [10]:
```
from shapely.geometry import Point

# Create GeoDataframe
df1["Coordinates"] = list(zip(df1.longitude, df1.latitude))
df1["Coordinates"] = df1["Coordinates"].apply(Point)

gdf = gpd.GeoDataFrame(df1, geometry="Coordinates")
gdf = gdf.set_crs(epsg=4326)
```

In [11]:
```
# Create Map
zcta_df = gpd.read_file("https://www2.census.gov/geo/tiger/GENZ2018/shp/cb_2018_us_zcta510_500k.zip")
tract_df = gpd.read_file("https://www2.census.gov/geo/tiger/GENZ2018/shp/cb_2018_36_tract_500k.zip")

borough_map = {
    "005": "Bronx",
    "047": "Brooklyn",
    "061": "Manhattan",
    "081": "Queens",
    "085": "Staten Island"
}

tract_df["COUNTYFP"] = tract_df["COUNTYFP"].astype(str)
tract_df["borough"] = tract_df["COUNTYFP"].map(borough_map)

nyc_tracts = tract_df[tract_df["borough"].notna()].copy()

borough_df = nyc_tracts.dissolve(by="borough").reset_index()

target_crs = "EPSG:2263"

gdf = gdf.to_crs(target_crs)
zcta_df = zcta_df.to_crs(target_crs)
borough_df = borough_df.to_crs(target_crs)

nyc_union = borough_df.union_all()
nyc_zcta = zcta_df[zcta_df.intersects(nyc_union)].copy()

gdf_zip = gpd.sjoin(gdf, nyc_zcta, how="left", predicate="within")
```

2.2.1 Map1: Price of Airbnb Listing (Outcome Variable)

In [31]:

```
map_plot = nyc_zcta_price.to_crs(epsg=4326)
borough_plot = borough_df.to_crs(epsg=4326)

fig, ax = plt.subplots(1,1,figsize=(10,10))

map_plot.plot(
    column="price",
    cmap="Blues",
    linewidth=0.5,
    edgecolor="gray",
    legend=True,
    ax=ax)

borough_plot.boundary.plot(ax=ax, color="black", linewidth=2)

for _, row in borough_plot.iterrows():
    p = row.geometry.representative_point()
    ax.text(
        p.x,
        p.y,
        row["borough"],
        ha="center",
        va="center",
        fontsize=11,
        fontweight="bold",
        color="black")

ax.set_title("Average Airbnb Listing Price by ZIP Code")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")
ax.set_aspect("auto")

plt.show()
```

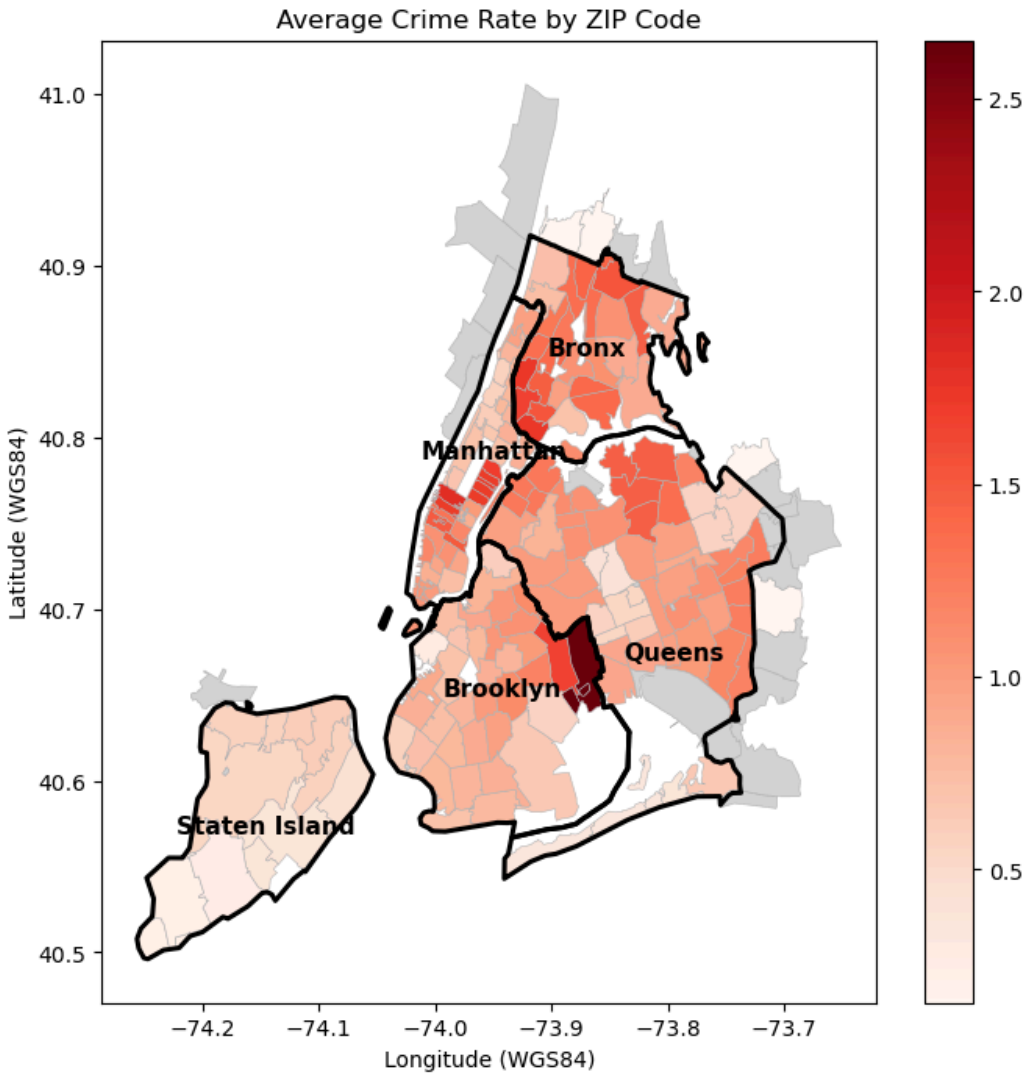


The first map shows the spatial distribution of ***average Airbnb listing prices*** across ZIP codes in New York City. High prices are strongly concentrated in Manhattan, particularly in central and southern areas, which appear darker on the map. Some areas in western Brooklyn and parts of Queens also display moderately high prices, although they remain generally lower than those in Manhattan. In contrast, most ZIP codes in the Bronx and Staten Island show much lighter colors, indicating lower average Airbnb prices. This pattern is consistent with the stronger economic environment and tourism activity in Manhattan, which likely increases visitors' willingness to pay for short-term accommodation. A few ZIP codes appear noticeably darker than surrounding areas, including one location in Queens and a small area in Staten Island. These may represent local outliers with particularly high prices, possibly reflecting strong tourism demand or luxury listings.

2.2.2 Map 2: Crime Rate (Important Independent Variable)

In [37]:
```
map_plot = nyc_zcta_crime.to_crs(epsg=4326)
borough_plot = borough_df.to_crs(epsg=4326)

fig, ax = plt.subplots(figsize=(8, 8))

map_plot.plot(
    ax=ax,
    column="crime_rate",
    cmap="Reds",
    edgecolor="0.7",
    linewidth=0.4,
    legend=True,
```

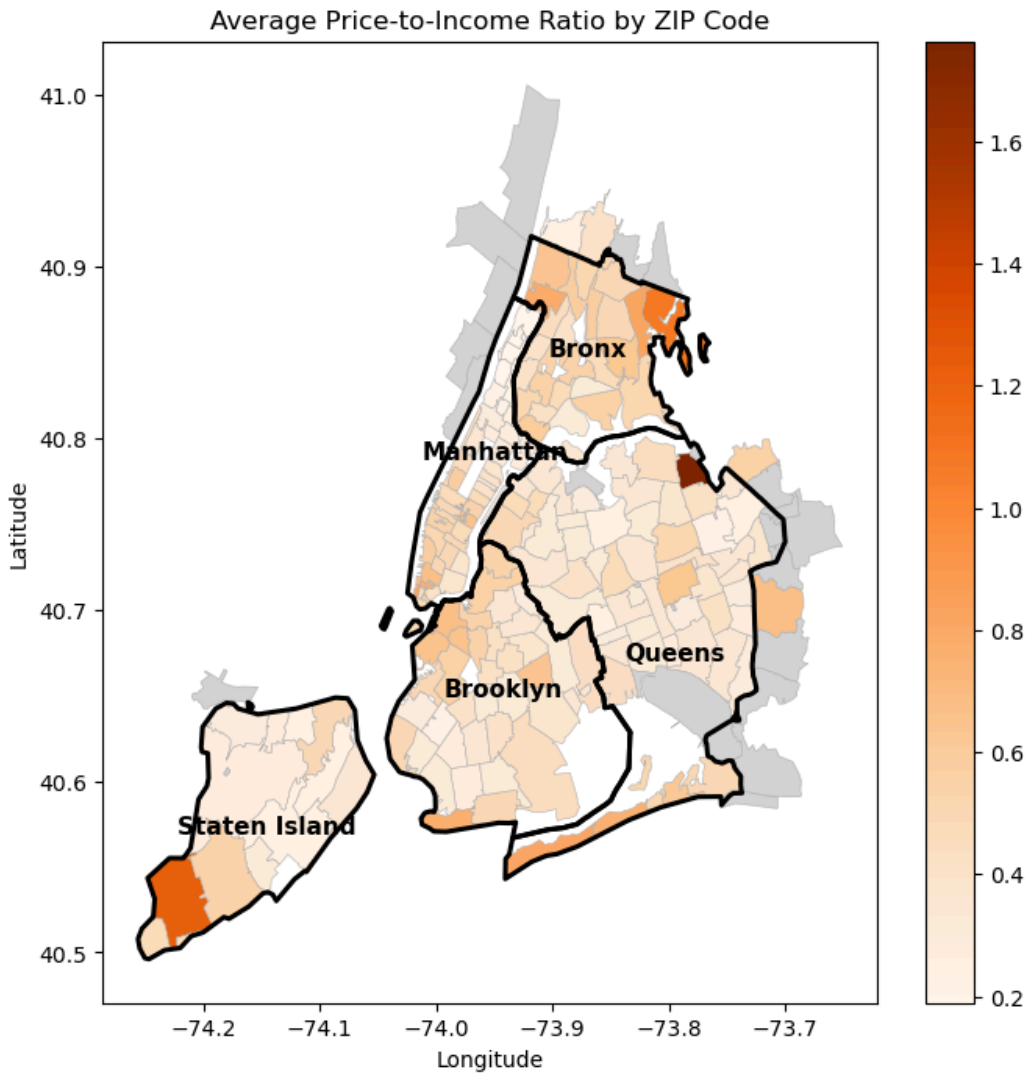
```
    missing_kwds={
        "color": "lightgrey",
        "label": "No data"})

borough_plot.plot(ax=ax, facecolor="none", edgecolor="black", linewidth=2.2,)

for _, row in borough_plot.iterrows():
    p = row.geometry.representative_point()
    ax.text(
        p.x, p.y,
        row["borough"],
        ha="center",
        va="center",
        fontsize=11,
        fontweight="bold",
        color="black")

ax.set_title("Average Crime Rate by ZIP Code")
ax.set_xlabel("Longitude (WGS84)")
ax.set_ylabel("Latitude (WGS84)")
ax.set_aspect("auto")

plt.show()
```



The second map shows the spatial distribution of ***relative crime rates*** across ZIP codes in New York City. Crime rates are normalized by the citywide average, where values above 1 indicate higher-than-average crime levels. Several areas in the Bronx and parts of central Brooklyn display crime rates above the city average. Some ZIP codes in central Manhattan also show slightly elevated crime levels, although overall crime rates in Manhattan appear more moderate compared with the Bronx. Comparing this map with the Airbnb price map suggests that areas with lower crime rates often correspond to higher Airbnb prices. For example, several areas in the Bronx and central Brooklyn exhibit relatively high crime levels and lower Airbnb prices. A few ZIP codes in central Brooklyn appear significantly darker than surrounding areas, indicating potential crime hotspots that stand out as spatial outliers.

2.2.3 Map 3: Price to Income Ratio

In [36]:
```
map_plot = nyc_zcta_map.to_crs(epsg=4326)
borough_plot = borough_df.to_crs(epsg=4326)

fig, ax = plt.subplots(figsize=(8, 8))

map_plot.plot(
    ax=ax,
    column="Price_to_Income Ratio",
    cmap="Oranges",
    edgecolor="0.7",
    linewidth=0.4,
    legend=True,
    missing_kwds={
        "color": "lightgrey",
        "label": "No data"})

borough_plot.plot(ax=ax, facecolor="none", edgecolor="black", linewidth=2.2,)

for _, row in borough_plot.iterrows():
    p = row.geometry.representative_point()
    ax.text(
        p.x, p.y,
        row["borough"],
        ha="center",
        va="center",
        fontsize=11,
        fontweight="bold",
        color="black")

ax.set_title("Average Price-to-Income Ratio by ZIP Code")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")
ax.set_aspect("auto")

plt.show()
```

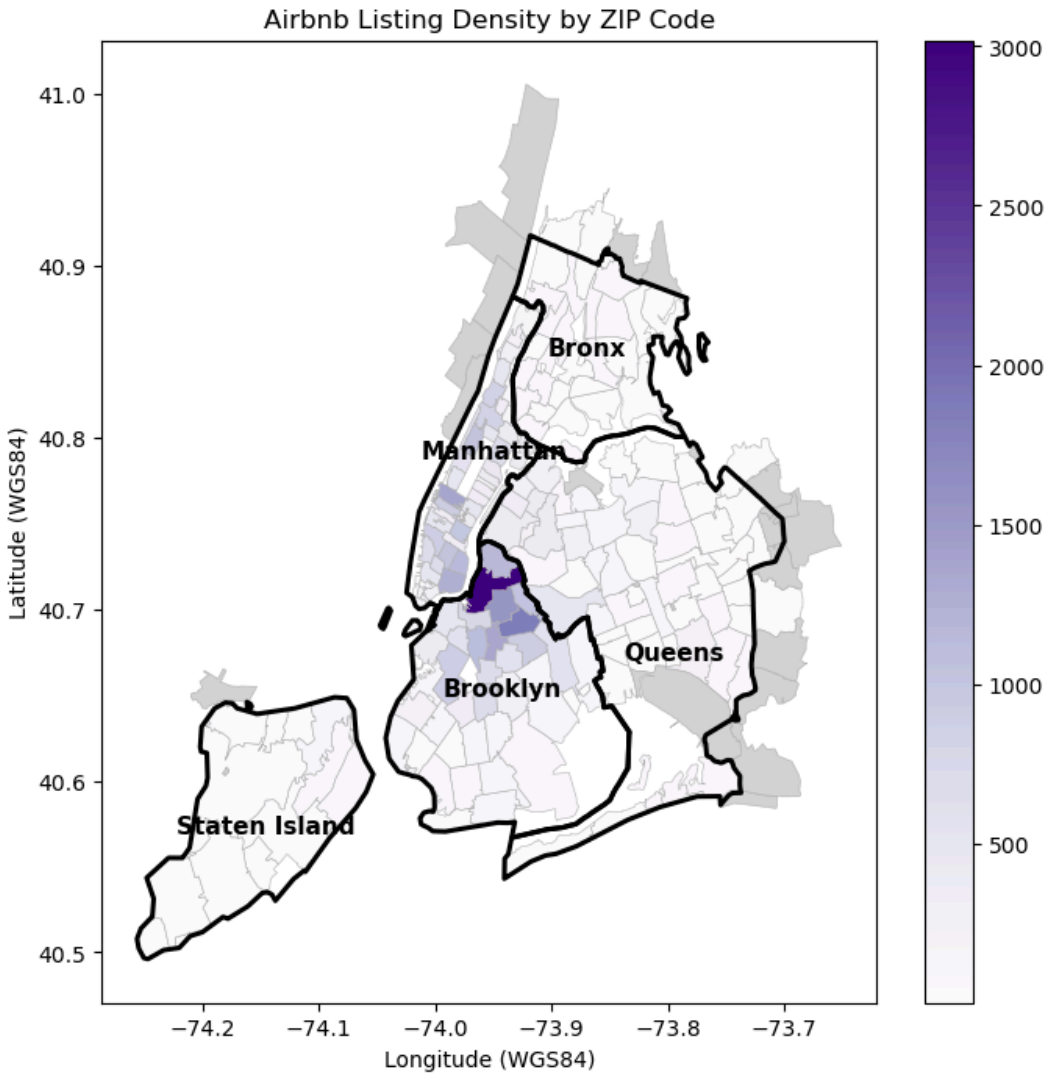


The third map shows the spatial distribution of the ***average price-to-income ratio*** across ZIP codes. This ratio measures how expensive Airbnb listings are relative to local income levels. Several areas in the Bronx display relatively high ratios. Although Airbnb prices in these areas are not the highest in the city, local income levels are lower, which results in higher price-to-income ratios. In contrast, many parts of Manhattan show more moderate ratios despite relatively high Airbnb prices. This likely reflects the much higher income levels in Manhattan, which reduce the relative burden of accommodation prices. Some coastal ZIP codes in Staten Island and Queens appear noticeably darker than surrounding areas, suggesting unusually high price-to-income ratios that may represent local outliers.

2.2.4 Map 4: Airbnb Listing Density

In [38]:
```
zip_listings = (
    gdf_zip
    .groupby("ZCTA5CE10")
    .size()
    .reset_index(name="listing_count"))

nyc_zcta_listings = nyc_zcta.merge(zip_listings, on="ZCTA5CE10", how="left")

map_plot = nyc_zcta_listings.to_crs(epsg=4326)
borough_plot = borough_df.to_crs(epsg=4326)

fig, ax = plt.subplots(figsize=(8, 8))

map_plot.plot(
    ax=ax,
    column="listing_count",
    cmap="Purples",
    edgecolor="0.7",
    linewidth=0.4,
    legend=True,
```

```
    missing_kwds={
        "color": "lightgrey",
        "label": "No data"})

borough_plot.plot(ax=ax, facecolor="none", edgecolor="black", linewidth=2.2)

for _, row in borough_plot.iterrows():
    p = row.geometry.representative_point()
    ax.text(
        p.x, p.y,
        row["borough"],
        ha="center",
        va="center",
        fontsize=11,
        fontweight="bold",
        color="black")

ax.set_title("Airbnb Listing Density by ZIP Code")
ax.set_xlabel("Longitude (WGS84)")
ax.set_ylabel("Latitude (WGS84)")
ax.set_aspect("auto")

plt.show()
```



The fourth map illustrates the spatial distribution of Airbnb listing density across ZIP codes in New York City. Listing density measures the number of Airbnb listings in each ZIP code and reflects the intensity of Airbnb market activity. Listings are highly concentrated in Manhattan and Brooklyn. Several ZIP codes in central Brooklyn display particularly high listing density, while Manhattan also shows a strong concentration of Airbnb activity. These areas are close to major tourist attractions and the city center, which likely increases demand for short-term accommodation. In contrast, most ZIP codes in the Bronx and Staten Island show very low listing density, indicating that Airbnb activity is much less common in these areas.

2.3 Regressions

2.3.1 Intuition

The previous sections show substantial spatial variation in Airbnb prices and neighborhood conditions across New York City. In particular, areas with stronger economic environments, such as higher income levels and lower crime rates, tend to have higher Airbnb prices. While these patterns provide useful intuition, they do not formally test the relationship between various neighborhood economic conditions and Airbnb prices. To examine this relationship more systematically, this section estimates a series of ordinary least squares (OLS) regressions.

Economic intuition suggests that the relationship between Airbnb prices and local economic conditions is likely non-linear. When economic conditions improve, accommodation prices typically adjust proportionally rather than by a fixed dollar amount. This pattern is consistent with the data, where listing prices are highly dispersed and right-skewed. Taking the logarithm of price reduces the influence of extreme values and allows coefficients to be interpreted as approximate percentage changes.

For this reason, the main specification uses the logarithm of listing prices as the dependent variable. However, alternative regressions using price levels are also estimated to examine the robustness of the results.

2.3.2 Choice of Variables

The explanatory variables are chosen based on existing research on housing markets and Airbnb pricing. The regressions include variables capturing neighborhood economic conditions, short-term rental supply, and listing characteristics.

Neighborhood conditions are measured using average household income, GDP per capita, population density, and crime rate. Income and GDP reflect local economic strength, population density captures urban intensity and housing demand, and crime rate serves as a proxy for neighborhood safety.

The models also include supply and listing characteristics such as availability, minimum nights, room type, and number of reviews. These variables reflect hosting strategies, accommodation types, and listing popularity, all of which may influence pricing decisions.

2.3.3 Regression Table

The regressions examine how different factors affect Airbnb prices across New York City. The models gradually add groups of variables to observe how estimated relationships change when additional controls are included. This approach helps identify the effects of neighborhood conditions, listing characteristics, and location.

In [30]:
```
import statsmodels.api as sm
```

In [31]:
```
# dependent variables (outcome)
y = df1["price"]
y2 = df1["log_price"]

# create dummy variables
room_dummies = pd.get_dummies(df1["room_type"], drop_first=True).astype(float)
borough_dummies = pd.get_dummies(df1["neighbourhood_group"], drop_first=True).astype(float)
```

In [14]:
```
# Benchmark: price on crime_rate
Xb = sm.add_constant(df1[["crime_rate"]])
bench = sm.OLS(y, Xb, missing="drop").fit()

# Model 1: crime + income
X1 = sm.add_constant(df1[["crime_rate", "avg_income_10k"]])
reg1 = sm.OLS(y, X1, missing="drop").fit()

# Model 2: + GDP
X2 = sm.add_constant(df1[["crime_rate", "avg_income_10k", "gdp_10k"]])
reg2 = sm.OLS(y, X2, missing="drop").fit()

# Model 3: + room type dummies
X3 = pd.concat([df1[["crime_rate", "avg_income_10k", "gdp_10k"]], room_dummies], axis=1)
X3 = sm.add_constant(X3)
```

```
reg3 = sm.OLS(y, X3, missing="drop").fit()

# Model 4: + borough dummies
X4 = pd.concat([df1[["crime_rate", "avg_income_10k", "gdp_10k"]], room_dummies, borough_dummies], axis=1)
X4 = sm.add_constant(X4)
reg4 = sm.OLS(y, X4, missing="drop").fit()

# Stargazer Table 1
stargazer1 = Stargazer([bench, reg1, reg2, reg3, reg4])
stargazer1.title("Table 1: Neighborhood, Location and Housing Type Effects on Airbnb Prices")
stargazer1.custom_columns(["Benchmark","M1","M2","M3","M4"], [1,1,1,1,1])
display(HTML(stargazer1.render_html()))
```

Table 1: Neighborhood, Location and Housing Type Effects on Airbnb Prices

					Dependent variable: price
	Benchmark	M1	M2	M3	M4
	(1)	(2)	(3)	(4)	(5)
Brooklyn					-20.007***
					(5.834)
Manhattan					-3.519***
					(0.552)
Private room				-111.748***	-111.309***
				(2.141)	(2.144)
Queens					-30.233***
					(6.167)
Shared room				-135.132***	-134.355***
				(6.921)	(6.923)
Staten Island					-41.088***
					(9.140)
avg_income_10k		10.649***	11.035***	6.670***	11.943***
		(0.294)	(1.890)	(1.839)	(0.976)
const	127.118***	1.287	-2.262	102.415***	77.515***
	(3.030)	(4.580)	(17.792)	(17.403)	(11.746)
crime_rate	25.603***	13.200***	13.313***	11.951***	12.349***
	(2.829)	(2.812)	(2.865)	(2.786)	(2.794)
gdp_10k			-0.061	0.233	-0.953***
			(0.295)	(0.286)	(0.191)
Observations	48895	48895	48895	48895	48895
R^2	0.002	0.028	0.028	0.082	0.082
Adjusted R^2	0.002	0.028	0.028	0.082	0.082
Residual Std. Error	239.956 (df=48893)	236.795 (df=48892)	236.797 (df=48891)	230.146 (df=48889)	230.119 (df=48887)
F Statistic	81.925*** (df=1; 48893)	699.557*** (df=2; 48892)	466.376*** (df=3; 48891)	869.998*** (df=5; 48889)	623.510*** (df=7; 48887)
Note:					*p<0.1; **p<0.05; ***p<0.01

In [32]:
```
# Model 5: log(price) on crime + income
X5 = sm.add_constant(df1[["crime_rate", "avg_income_10k"]])
reg5 = sm.OLS(y2, X5, missing="drop").fit()
```

```
# Model 6: + listing characteristics + reviews
X6 = sm.add_constant(df1[["crime_rate", "avg_income_10k", "availability_365", "minimum_nights", "number_of_r
reg6 = sm.OLS(y2, X6, missing="drop").fit()

# Model 7: + room type
X7 = pd.concat([df1[["crime_rate","avg_income_10k","availability_365","minimum_nights","number_of_reviews"]]
reg7 = sm.OLS(y2, X7, missing="drop").fit()

# Model 8: + borough + room type
X8 = pd.concat([df1[["crime_rate","avg_income_10k","availability_365","minimum_nights","number_of_reviews"]]
X8 = sm.add_constant(X8)
reg8 = sm.OLS(y2, X8, missing="drop").fit()

# Stargazer Table 2
stargazer2 = Stargazer([reg5, reg6, reg7, reg8])
stargazer2.title("Table 2: Airbnb Market Characteristics and Log Prices")
stargazer2.custom_columns(["M5","M6","M7","M8"], [1,1,1,1])
display(HTML(stargazer2.render_html()))
```

Table 2: Airbnb Market Characteristics and Log Prices

				Dependent variable: log_price
	M5	M6	M7	M8
	(1)	(2)	(3)	(4)
Brooklyn				-0.389***
				(0.011)
Manhattan				-1.245***
				(0.008)
Private room			-0.117***	-0.776***
			(0.009)	(0.005)
Queens				-0.479***
				(0.013)
Shared room			-0.707***	-1.170***
			(0.032)	(0.015)
Staten Island				-0.855***
				(0.025)
availability_365		0.001***	0.002***	0.001***
		(0.000)	(0.000)	(0.000)
avg_income_10k	0.066***	0.066***	0.246***	0.165***
	(0.001)	(0.001)	(0.001)	(0.001)
const	3.824***	3.791***		3.664***
	(0.013)	(0.013)		(0.025)
crime_rate	0.055***	0.037***	1.184***	0.036***
	(0.008)	(0.008)	(0.011)	(0.006)
minimum_nights		-0.000**	-0.000	-0.002***
		(0.000)	(0.000)	(0.000)
number_of_reviews		-0.001***	0.002***	-0.001***
		(0.000)	(0.000)	(0.000)
Observations	48895	48895	48895	48895
R^2	0.125	0.138	0.950	0.469
Adjusted R^2	0.125	0.138	0.950	0.469
Residual Std. Error	0.650 (df=48892)	0.646 (df=48889)	1.074 (df=48888)	0.507 (df=48884)
F Statistic	3505.965*** (df=2; 48892)	1563.782*** (df=5; 48889)	131693.224*** (df=7; 48888)	4322.158*** (df=10; 48884)
Note:				*p<0.1; **p<0.05; ***p<0.01

2.3.4 Preferred Model Specification

Among all specifications, the preferred model is Model 8 in Table 2, which uses the logarithm of price as the dependent variable and includes neighborhood economic variables, listing characteristics, room type, and borough indicators.

The preferred regression specification can be written as:

$$\log(Price_i) = \beta_0 + \beta_1 CrimeRate_i + \beta_2 Income_i \tag{1}$$

$$+\beta_3 Availability_i + \beta_4 MinimumNights_i + \beta_5 Reviews_i \tag{2}$$

$$+\beta_6 PrivateRoom_i + \beta_7 SharedRoom_i \tag{3}$$

$$+\beta_8 Brooklyn_i + \beta_9 Manhattan_i + \beta_{10} Queens_i + \beta_{11} StatenIsland_i + \varepsilon_i \tag{4}$$

This specification is preferred because it includes the most comprehensive set of controls, incorporating neighborhood economic conditions, listing characteristics, room type, and borough indicators. Compared with earlier models, Model 8 achieves a reasonably high R^2, indicating that the model explains a substantial share of the variation in Airbnb prices. The adjusted R^2 remains at a similar level, suggesting that the improvement in model fit is not simply driven by adding more explanatory variables. In addition, the residual standard error decreases relative to simpler specifications, which indicates a better overall fit of the model.

Using the logarithm of price also has practical advantages. The log specification allows coefficients to be interpreted approximately as percentage changes, making it easier to compare the relative importance of different factors. It also reduces the influence of extremely expensive listings, which are common in Airbnb data.

2.3.5 Regression Results

The regression results show how neighborhood economic conditions and listing characteristics are related to Airbnb prices. Focusing on the preferred specification (Model 8), we can see several patterns emerge.

First, the coefficient on average income is positive and statistically significant, suggesting that listings located in higher-income neighborhoods tend to have higher prices. This is consistent with the idea that wealthier areas may attract stronger tourism demand and offer better amenities. Second, housing characteristics play an important role. Compared with entire homes, both private rooms and shared rooms are associated with significantly lower prices, reflecting differences in space and privacy. Finally, listing characteristics and location also contribute to price variation. Variables such as availability, minimum nights, and the number of reviews are related to pricing outcomes, while borough indicators capture remaining location differences across the city.

2.4 Conclusion

This project examines how neighborhood economic conditions are associated with Airbnb price levels in New York City, with housing market conditions serving as a potential mechanism linking local economic environments to short-term rental prices. Using the NYC Airbnb 2019 dataset combined with borough-level economic indicators and neighborhood characteristics, the analysis first documents descriptive patterns in prices and neighborhood conditions and then evaluates these relationships more formally through regression analysis.

The descriptive analysis reveals substantial spatial heterogeneity in Airbnb prices across boroughs. Higher prices are concentrated in areas with stronger economic environments and more active housing markets, particularly in Manhattan and parts of Brooklyn. Differences in listing composition also play an important role. Boroughs with a higher share of entire-home listings tend to exhibit higher price levels, reflecting differences in the types of housing products supplied in the short-term rental market.

The regression analysis provides further support for these patterns. Listings located in economically stronger neighborhoods, particularly those with higher income levels, tend to have higher prices. Housing characteristics such as listing type also have a large impact on prices, while borough differences continue to reflect broader variation in housing demand and neighborhood conditions.

Overall, the findings suggest that local economic environments are closely linked to Airbnb price levels through housing market demand and supply conditions. Although the analysis shows clear associations, future research could further explore the potential causal relationship between short-term rental activity and local housing markets.

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Data Source Appendix

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